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# Cubosomes with pH-Triggered Cubic Phase Transition Enable Cytosolic mRNA Delivery for Acute Respiratory Distress Syndrome Therapy

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## Abstract

Interleukin-10 (IL-10) represents a promising anti-inflammatory candidate for acute respiratory distress syndrome (ARDS) treatment, but its clinical application is hindered by short half-life and poor lung exposure when delivered as recombinant protein. We developed virus-mimetic cubosomes (VE-CB) based on vitamin E-derived ionizable lipid and glyceryl monooleate for cytosolic delivery of IL-10 mRNA. The system features a pH-triggered lamellar vesicle-to-bicontinuous cubic phase transition that mimics viral membrane fusion mechanisms. Upon acidification to endosomal pH (5.0-5.5), VE-CB underwent a rapid lamellar vesicle-to-bicontinuous cubic phase transition, as confirmed by small-angle X-ray scattering and cryo-transmission electron microscope. This structural switch enhanced membrane fluidity and fusion capability, enabling efficient endosomal escape. In ARDS mice, intranasal administration of mIL-10@VE-CB achieved sustained pulmonary IL-10 expression (462 pg/mL), significantly reduced inflammatory cell infiltration, decreased pro-inflammatory cytokines, and restored macrophage efferocytosis. The virus-mimetic cubosome platform represents a safe, efficient, and versatile strategy for pulmonary mRNA delivery, offering substantial clinical translational potential for ARDS precision therapy through structure-enabled cytosolic delivery and synergistic anti-inflammatory mechanisms.

**Key words:** Acute respiratory distress syndrome; IL-10 mRNA; Cubosome; Cytosolic mRNA delivery; Endosomal escape

## 1. Introduction

Acute respiratory distress syndrome (ARDS) is a highly devastating critical respiratory disorder characterized by diffuse alveolar damage, intense pulmonary inflammatory responses, and progressive hypoxemic respiratory failure [1]. The pathological progression of ARDS is driven by the cytokine storm that disrupts the alveolar-endothelial barrier [2-4]. Anti-inflammatory cytokines, interleukin-10

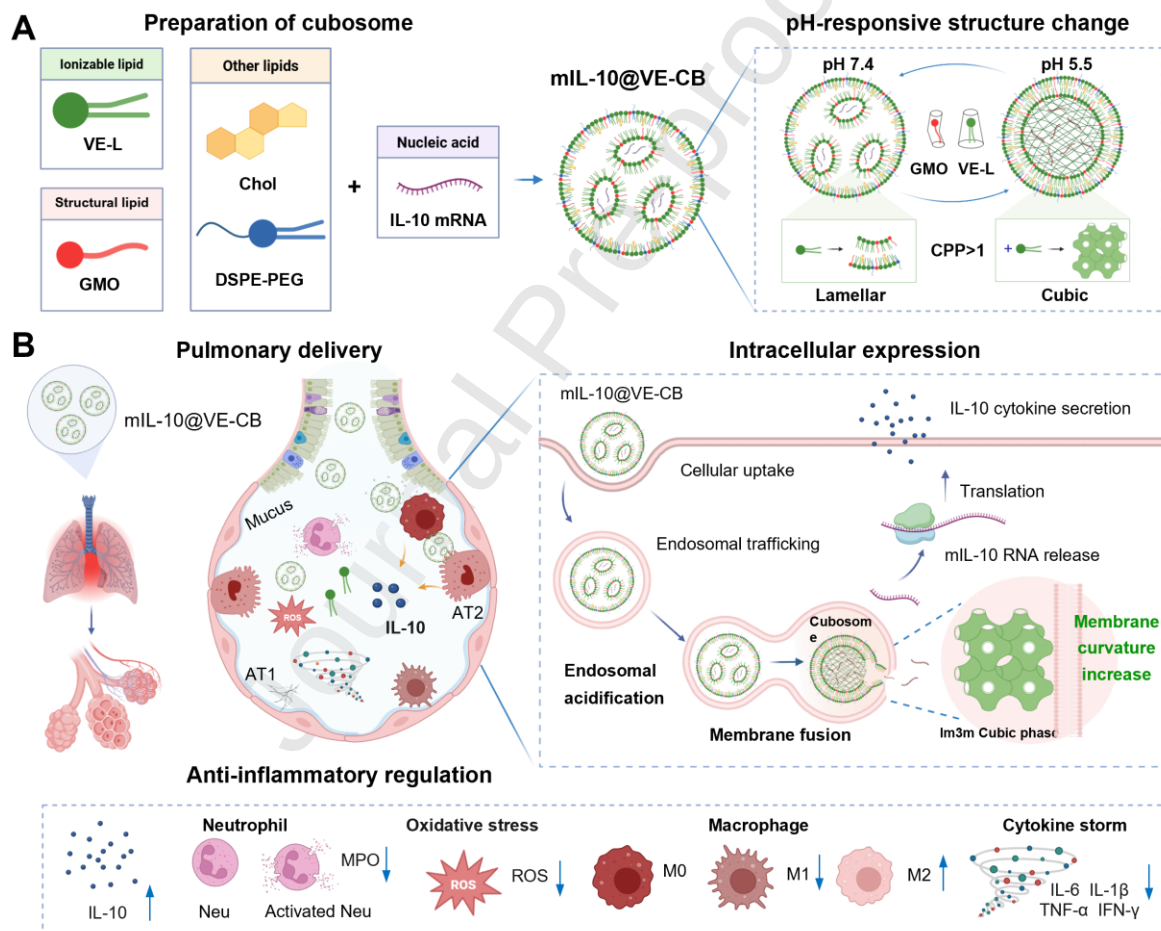
(IL-10), could simultaneously suppress TNF- $\alpha$ /IL-6 cascades, polarize macrophages toward the anti-inflammatory M2 phenotype and promote alveolar epithelial regeneration, making it the key molecule able to curb inflammation while restoring tissue integrity [5, 6]. However, recombinant IL-10 suffers from a short half-life and poor lung exposure [7-9], necessitating repeated high-dose injections that risk systemic immunosuppression [10]. Delivering mRNA encoding IL-10 therefore offers a precision intervention [11, 12]. Single dose could achieve rapid, local and sustained expression, providing the sustained anti-inflammatory and reparative signal ARDS urgently demands without compromising safety [13].

Conventional lipid nanoparticles (LNPs), often possessing lamellar vesicle structure or disordered internal core, have brought mRNA therapeutics to the clinic [14, 15], yet their efficacy is curtailed by a common bottleneck. It has been reported most endocytosed mRNA remains trapped in acidified endosomes and is subsequently degraded in lysosomes [16]. Since the topological structure of the carrier has a coupling mechanism with intracellular dynamics, improving their structure is of great significance for enhancing the efficiency of nucleic acid delivery [17, 18]. The influenza virus-derived membrane fusion system has been developed and proven to have good endosome escape ability [19-21]. Upon endosomal acidification (pH 5.0-5.5), viral hemagglutinin undergoes a conformational rearrangement that converts pre-organized high-curvature lattices into a fusion pore, allowing the encapsulated macromolecules to escape into the cytosol with high efficiency [22]. Therefore, endowing LNP with virus-like topological switches and acid-triggered membrane fusion, it is anticipated a step-change in cytosolic mRNA delivery that may pave the way for programmable nucleic-acid therapeutics [23].

Bicontinuous cubic-phase lipids formed cubosomes represent a unique class of soft mesoporous materials in which a curved lipid bilayer partitions 3D continuous water channels with pore diameters of 5-6 nm and negative Gaussian curvature throughout the  $Im3m$  or  $Pn3m$  space group [24, 25]. Unlike conventional liposomes or LNPs, these self-assembled  $Q_{II}$  phases possess a critical packing parameter ( $CPP > 1$ ), which is usually enforced by cone-shaped mono-acyl glycerols such as glyceryl monooleate (GMO) [26]. This yields membrane planes that can bend, buckle and reseal without losing long-range order [27, 28]. Recent cryo-transmission electron microscope (Cryo-TEM) and small-angle X-ray scattering (SAXS) studies reveal that the mean curvature  $H \approx 0$  topology of  $Q_{II}$  is topologically equivalent to the lattice found in influenza and coronavirus envelopes, where pre-organized negative Gaussian curvature primes the membrane for subsequent fusion events [29]. Harnessing this structure-driven elastic energy for cytosolic nucleic acid delivery, however, remains largely unexplored.

To integrate the physiological stability with the endosomal membrane fusion capacity [30], we engineered cubosomes (VE-CB) based on the vitamin E-derived ionizable lipid (VE-L). As shown in Fig. 1A, VE-CB could function as a binary-phase switch to enable on-demand endosomal release of mRNA and ARDS therapy. Composed of our synthesized ionizable lipid (VE-L) and GMO [31], VE-CB adopt a lamellar vesicle structure at pH 7.4, encapsulating and shielding mRNA in a similar manner to standard LNPs [32]. Upon entry into pH 5.0-5.5 endosomes, protonation of the tertiary amines in

VE-L expands the head-group area, triggering an instantaneous lamellar vesicle-to-bicontinuous cubic transition [33-35]. The resulting negative Gaussian curvature and topological defects drop the fusion-energy barrier, releasing mRNA into the cytosol, thus breaking the endosomal bottleneck that limits conventional lipid systems [36]. In this study, the ability of VE-CB to deliver multiple types of nucleic acids was assessed, with a focus on analyzing its mRNA delivery efficiency, membrane fusion capacity, cellular uptake mechanism, as well as its tissue distribution and therapeutic efficacy in lung delivery. VE-CB was successfully used to deliver IL-10 mRNA, and effectively alleviated lung inflammatory responses in an ARDS mouse model (Fig. 1B). Inspired by the pH-triggered membrane fusion mechanism of virus, we engineered the virus-mimetic cubosomes that functionally mimic this process. These results demonstrate favorable therapeutic efficacy and biosafety, fully confirming the unique advantages and clinical translational potential of the virus-mimetic cubosome system for ARDS treatment.



**Fig. 1. The therapeutic mechanism of mIL-10@VE-CB in ARDS treatment.** (A) Preparation of VE-CB via the ethanol injection method and pH-responsive structural transition. (B) Process and mechanism of mIL-10@VE-CB for ARDS treatment. Schematic diagram was created with BioRender.com.

## 2. Results and discussion

### 2.1 Physicochemical characterization and formulation optimization of VE-CB

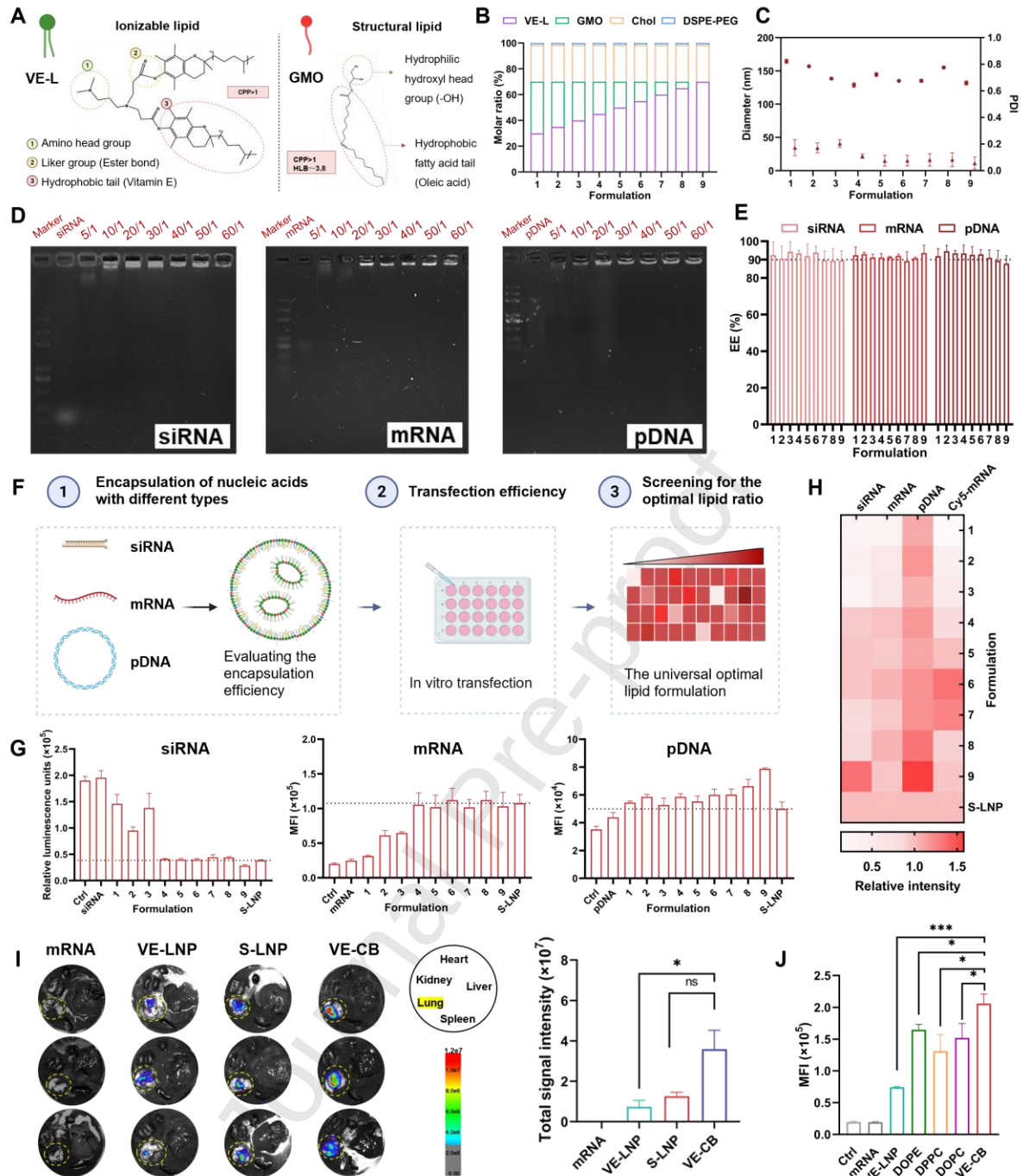
Glyceryl monooleate (GMO) is reported to have a unique amphiphilic molecular structure and a critical packing parameter greater than 1 ( $CPP > 1$ , enabling the spontaneous emergence of a bicontinuous cubic structure) [37, 38]. It has a hydrophilic-lipophilic balance (HLB) of approximately 3.8 to highlight its strong lipophilicity. In our previous study, VE-L combined high mRNA transfection potency with an antioxidant tail that scavenges excess ROS in inflamed lung tissue [31]. At pH 7.4, the neutral VE-L possesses a bulky vitamin E tail that gives the CPP already close to or above 1. Upon acidification to pH 5.5, protonation of the tertiary amines introduces positive charges on the headgroup. As displayed in Fig. 2A, electrostatic repulsion expands the effective headgroup area, while enhanced tail repulsion further increases the CPP, driving a transition from a lamellar vesicle to a cubic phase with negative Gaussian curvature [39]. We therefore selected VE-L as the pH-responsive cornerstone for cubosome assembly with GMO. To engineer VE-L-derived virus-mimetic cubosomes (VE-CB) that combined efficient transfection potency, stable nucleic acid encapsulation, and acid-triggered phase transition, we conducted a systematic formulation screening (Fig. 2B). Cholesterol (Chol, 28.5%) and PEGylated lipid (DSPE-PEG<sub>2000</sub>, 1.5%) were kept constant [40, 41], while the molar proportions of VE-L (30%-70%) and GMO (0%-40%) were varied. All compositions yielded VE-CB with uniform size ( $143.5 \pm 12.3$  nm,  $PDI < 0.2$ ) and neutral surface charge at pH 7.4. When the pH decreased to 6.5 or 5.5, the zeta potential increased gradually (Fig. 2C and S1A). This pH-dependent charge shift is attributed to the protonation of tertiary amines in VE-L, which facilitates interaction with endosomal membranes [42]. Furthermore, agarose-gel electrophoresis assays [43] revealed complete encapsulation of Luc siRNA (21 bp), EGFP mRNA (1007 nt), and EGFP pDNA (4700 bp) at lipid/nucleic acid mass ratios (L/N ratio)  $\geq 10:1$  (Fig. 2D). Consequently, L/N ratio of 20:1 was adopted, at which all 9 formulations achieved  $> 90\%$  encapsulation efficiency for pDNA, mRNA and siRNA (Fig. 2E). The decrease in zeta potential also confirmed the effective encapsulation at this ratio (Fig. S1B).

To identify the composition conferring universal nucleic acid delivery capability, 9 formulations were benchmarked against SM102-based LNP (S-LNP, clinical standard) [44] and GMO-free VE-L-based LNP (VE-LNP, negative control) for their ability to transfect Luc siRNA, EGFP mRNA, and EGFP pDNA *in vitro*, while parallel flow cytometry (FCM) quantified Cy5-mRNA uptake mediated by different formulations (Fig. 2F). The transfection efficiency and uptake results were shown in Fig. 2G and S1C. Performance metrics were visualized into a heatmap referenced to S-LNP (darker red indicating better performance, Fig. 2H). Formulations 4-9 all exhibited robust transfection and cellular uptake. Among them, the VE-L: GMO ratio at 55: 15 (formulation 6) yielded siRNA and mRNA efficiencies equivalent to S-LNP and superior pDNA expression, reflecting an enhanced capacity for large-cargo delivery. Therefore, the optimal formulation was identified as VE-L: GMO: Chol: DSPE-PEG<sub>2000</sub> at a molar ratio of 55: 15: 28.5: 1.5 (mol%), with a L/N mass ratio of 20: 1. VE-CB exhibited a uniform particle size distribution and excellent stability (Fig. S1D), with a determined apparent dissociation constant ( $pK_a$ ) of 6.09 (Fig. S1E). This  $pK_a$  was well-matched to the pulmonary mucus microenvironment, endowing VE-CB with electroneutral characteristics in mucus ensuring efficient

mucus penetration ability [45]. Transmission electron microscope (TEM) and cryogenic TEM (Cryo-TEM) were utilized for the intuitive characterization of morphological features. In Fig. S1F and S1G, VE-CB exhibited a uniform spherical morphology, and the experimentally observed particle size matched the DLS-detected results. To directly compare the *in vivo* performance of the selected VE-CB with the clinical-grade S-LNP, we further employed luciferase mRNA (Luc mRNA) as a reporter gene and administered mLuc@VE-CB to mice via the intranasal route (*i.n.*) and luciferase activity of lung was quantified *ex vivo*. Compared with mLuc@VE-LNP and mLuc@S-LNP at the same dose, the mLuc@VE-CB group exhibited significantly higher luciferase activity (Fig. 2I). These results demonstrated that VE-CB possessed superior transfection efficiency for local mRNA delivery, providing key *in vivo* evidence for its clinical translation.

To rigorously establish GMO as an indispensable structural element of VE-CB, as displayed in Fig. S1H, we substituted it with clinically prevalent helper lipids (DOPE, DPPC, DOPC) [46, 47], while keeping all other components constant, and tested their transfection efficiency. The decreased mean fluorescence intensity (MFI) in other groups demonstrated that GMO was essential for initiating and stabilizing the cubic architecture, whose synergy with VE-L may enable the topological switch that driven efficient endosomal escape (Fig. 2J).





**Fig. 2. Systematic screening of cubosome formulations.** (A) Structures of GMO and VE-L (created with BioRender.com). (B) Cubosome formulations with varied molar ratios of VE-L to GMO (VE-L: 30%-70%, GMO: 0%-40%). (C) Particle size and PDI of different formulations (n = 3). (D) At different mass ratios of L/N, agarose gel electrophoresis analysis of nucleic acid encapsulation, including three types of nucleic acids: Luc siRNA, EGFP mRNA, and EGFP pDNA. (E) Encapsulation efficiency for different nucleic acids (n = 3). (F) Schematic diagram for screening the optimal formulation (created with BioRender.com). (G) Transfection efficiency for different nucleic acids by 9 formulations, quantified by FCM (n = 3). (H) Heatmap integrating transfection efficiency and cellular uptake of 9 formulations, with commercial S-LNP as the reference. The depth of red indicates performance level (the darker red, the better performance). (I) Bioluminescence imaging and quantitative analysis of

luciferase expression in major organs of mice intranasally (*i.n.*) administered mLuc@VE-CB, mLuc@VE-LNP, or mLuc@S-LNP ( $n = 3$ ). (J) Transfection efficiency of VE-CB and formulations with GMO replaced by clinically common helper lipids was determined by FCM ( $n = 3$ ). Data were represented as mean  $\pm$  SD. Statistical analysis was conducted using two-tailed unpaired Student's *t*-test.  $*p < 0.05$ ,  $***p < 0.001$ , *ns* means no significance.

## 2.2 The pH-responsive membrane-fusion behavior of VE-CB

We first conducted a systematic characterization of the cellular uptake behavior of VE-CB. FCM and confocal laser scanning microscope (CLSM) results consistently demonstrated that VE-CB exhibited significantly higher internalization efficiency in MLE-12 cells compared to VE-LNP (Fig. S2A and S2B), with a concurrent increase in mRNA delivery into the cells (Fig. 3A and 3B), indicating a strong correlation between carrier structure and nucleic acid delivery efficiency. To elucidate the uptake mechanism of VE-CB, we pre-treated MLE-12 cells with a panel of endocytosis inhibitors [48], and subsequently calculated the cellular uptake inhibition rate (CUIR) from DiD@VE-CB mean fluorescence intensity (Fig. 3C). Incubation at 4°C led to 90% uptake inhibition rate, and pretreatment with M- $\beta$ -CD reduced the signal by  $\geq 60\%$ , confirming that VE-CB uptake was energy-dependent and primarily mediated by caveolin-mediated endocytosis [49]. Chloroquine also strongly suppressed the uptake of VE-CB, confirming that endosomal acidification is a common prerequisite for the internalization of nanoparticles. Besides, the addition of GMO did not alter the uptake pathway of the nanoparticles, and the uptake pattern of VE-CB was essentially consistent with that of VE-LNP (Fig. S2C). Subsequent endosomal escape experiments showed that the colocalization coefficient of Cy5-mRNA with lysosomes in the VE-CB group was significantly lower than that in the VE-LNP group (Fig. 3D and 3E), indicating a more efficient endosomal escape capacity [50, 51]. Collectively, VE-CB exploited caveolae-mediated cellular entry and rapidly exited acidic endosomal compartments, thereby boosting cytosolic mRNA delivery.

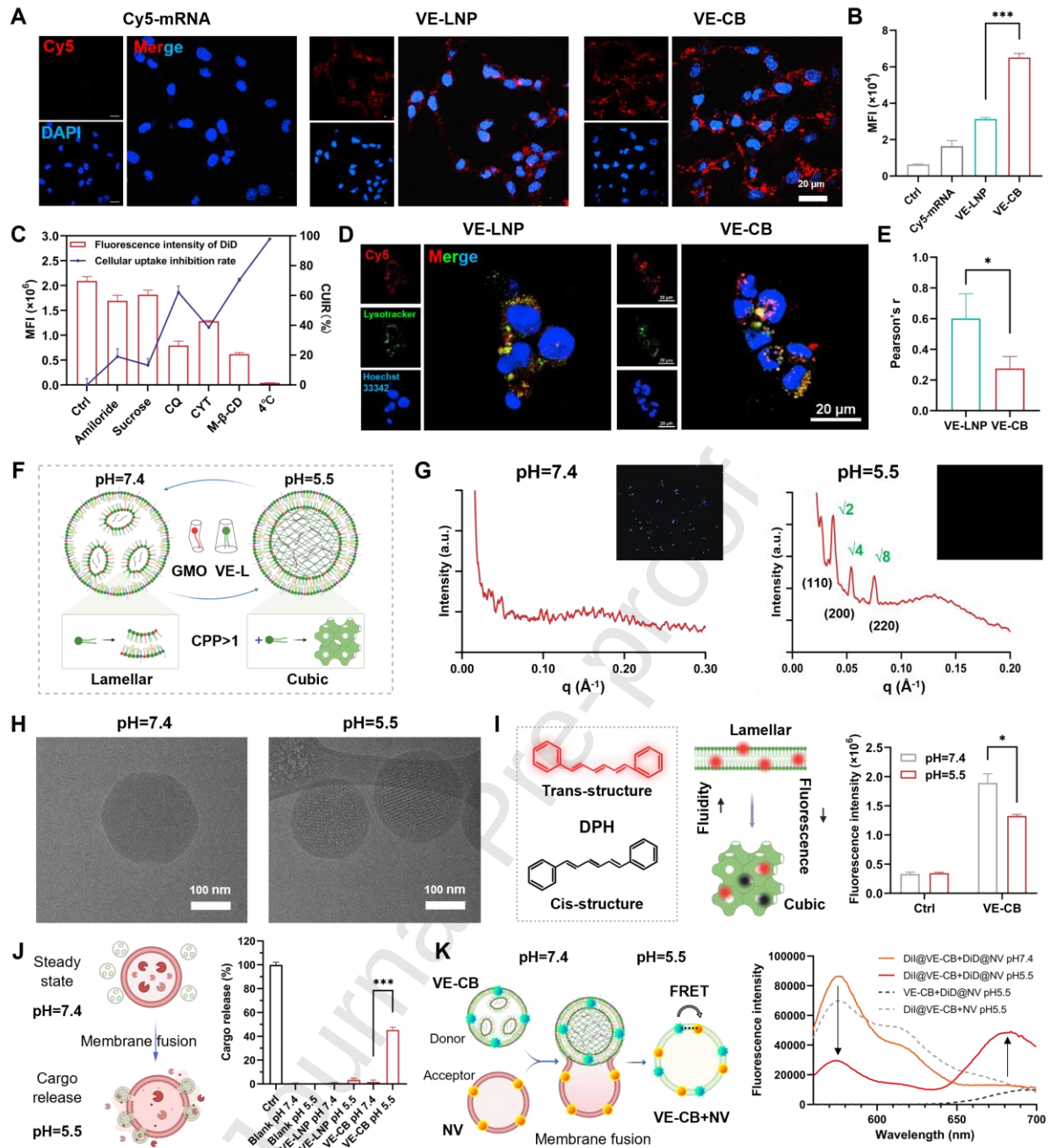
Given that carrier performance is structure-dependent, we believed that the high cellular uptake and robust endosomal escape of VE-CB was originated from its conformational switch in maturing endosomes (Fig. 3F). To explore whether the acidic endosomal environment induces structure exchange that enhances endosomal escape, small-angle X-ray scattering (SAXS) was first used to identify the structure of VE-CB. At pH 7.4, VE-CB did not exhibit characteristic diffraction peaks of the cubic phase; only broad and weak diffuse signals were observed, indicating that they predominantly exist as disordered lamellar vesicle structures [52], with no characteristic features of the cubic phase (Fig. 3G). Upon lowering the pH to 5.5, the zeta potential of VE-CB shifted to positive values (Fig. S2D). As shown in Table S1, the characteristic peaks of body-centered cubic (BCC) structures appeared [53]. At pH 5.5, the VE-CB exhibited distinct Bragg peaks at 0.037464, 0.053983, and 0.075813  $\text{\AA}^{-1}$ , with peak position ratios of  $\sqrt{2} : \sqrt{4} : \sqrt{8}$  corresponding to Miller indices (*hkl*) of (110), (200), and (220), respectively. These peaks were consistent with typical *Im3m* cubic phase structures [54-56]. Under polarized light microscope (PLM), VE-CB showed bright interference fringes at pH 7.4, whereas at pH



5.5 they yielded a dark and isotropic field, confirming the cubic phase transition under acidic conditions [57]. As shown in Fig. 3H, Cryo-TEM observations further validated this transition, displaying a typical bicontinuous 3D lattice architecture at pH 5.5 [23, 58]. It has been reported that the bicontinuous cubic phase exhibited higher turbidity than the lamellar vesicular phase in didodecyldimethylammonium bromide (DDAB)/phytantriol systems[28]. Analogously, in our system the significant rise of the OD<sub>400</sub> values for VE-CB at pH 5.5 provided independent optical evidence that the lamellar vesicles have transformed into cubic phases (Fig. S2E).

To probe whether VE-CB could exhibit virus-mimetic membrane fusion at endosomal pH, we conducted a comprehensive mechanistic study. Using 1, 6-diphenyl-1, 3, 5-hexatriene (DPH) as a probe [59], we quantified the membrane fluidity of VE-CB under different pH conditions. At pH 7.4, DPH fluorescence intensity was  $1.9 \pm 0.2 \times 10^6$ , while at pH 5.5, the signal decreased significantly, indicating enhanced membrane fluidity (Fig. 3I). This increase in membrane fluidity provided the dynamic possibility for subsequent fusion with the endosomal membrane. Additionally, the membrane destabilization ability of VE-CB was evaluated via a cargo release assay [60]. Red blood cell (RBC) membranes were used as endosome-mimicking models [61]. At pH 7.4, the cargo release rate of VE-CB group was less than 5%, while at pH 5.5 it rose to  $45.36 \pm 2.26\%$ , significantly higher than that in VE-LNP group (Fig. 3J). To further verify the endosomal membrane fusion ability, a fluorescence resonance energy transfer (FRET) assay was performed [62]. VE-CB were labeled with DiI as donor, and RBC-derived nanovesicles (NV) were labeled with DiD as acceptor (Fig. 3K). The clear FRET “trade-off” trend was observed, indicating that the distance between VE-CB and NV has shortened to the effective range of FRET. No FRET signal was detected in these controls, ruling out non-specific fluorescence increases due to spectral overlap or aggregation. Collectively, these results signify efficient membrane fusion between VE-CB and NV. To further validate this result, we repeated the FRET assay using the another FRET pair (DiO/DiI) and observed similar pH-dependent fusion (Fig. S2F).

Based on the above data, we can infer that VE-CB in acidic endosomes has morphological transformation and virus-bionic membrane fusion characteristics. As the organelle acidified from 7.4 to 5.0, VE-L underwent protonation, increasing positive-charge density and expanding the headgroup area. These alterations generated highly curved bilayers that collectively drove VE-CB to rearrange from a lamellar vesicle to a cubic phase. The resulting cubosomes could bind tightly to the negatively charged endosomal membrane and ultimately releases mRNA into the cytosol via membrane fusion.



**Fig. 3. Cellular uptake and pH-triggered membrane fusion property of VE-CB.** Cellular uptake of MLE-12 cells after 4 h incubation with free Cy5-mRNA, Cy5-mRNA@VE-LNP and Cy5-mRNA@VE-CB (mRNA: 1  $\mu$ g/mL), which was analyzed by (A) CLSM and (B) FCM. Cy5-mRNA: red; DAPI: blue (nucleus). Scale bar: 20  $\mu$ m. (C) The cellular uptake intensity and CUIR of VE-CB in MLE-12 cells after different treatments. (D) CLSM images of MLE-12 cells after 4 h incubation with Cy5-mRNA@VE-LNP and Cy5-mRNA@VE-CB (mRNA: 1  $\mu$ g/mL). Lysotracker (green, endosome); Cy5-mRNA (red); Hoechst 33342 (blue, nucleus). Scale bar: 20  $\mu$ m. (E) Pearson correlation coefficient used to measure the degree between lysosome (green) and Cy5-mRNA (red) fluorescence. (F) Schematic diagram of the structural exchange and escape mechanism of VE-CB in endosome (created with BioRender.com). (G) SAXS patterns and PLM images of VE-CB at pH 7.4 and pH 5.5. (H) Cryo-TEM images of VE-CB at pH 7.4 and pH 5.5. Scale bar: 100 nm. (I) Principle of VE-CB fluidity

detection using DPH (created with BioRender.com) and quantification of VE-CB membrane fluidity under different pH conditions by detecting fluorescence intensity at 430 nm ( $n = 3$ ). (J) Membrane fusion properties of VE-CB were evaluated via cargo release assay ( $n = 3$ ) (created with BioRender.com). (K) FRET assay of DiI-labeled VE-CB and DiD-labeled NV (created with BioRender.com). Data were represented as mean  $\pm$  SD. Statistical analysis was conducted using two-tailed unpaired Student's t-test.  $*p < 0.05$ ,  $***p < 0.001$ , *ns* means no significance.

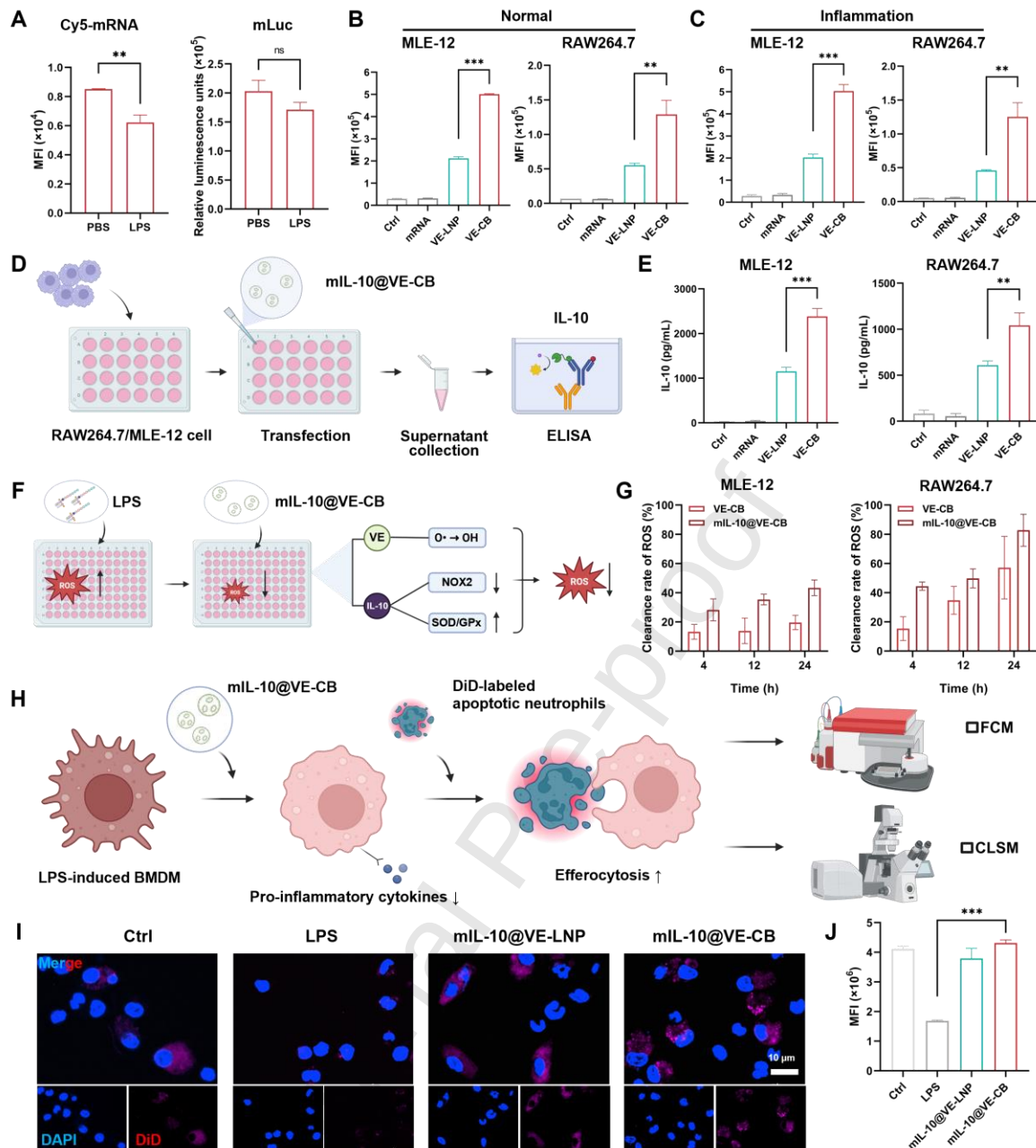
### 2.3 Regulation of VE-CB on cellular inflammation and efferocytosis effects

Prior to explore the effect of our system on cells, it was confirmed that VE-CB had excellent blood compatibility (within the detection range, the maximum safe concentration was 200  $\mu\text{g/mL}$ , Fig. S3A) and maintained good cytocompatibility in critical lung cells (Fig. S3B and S3C). During the progression of ARDS, the pulmonary microenvironment is dominated by a cytokine storm, and the transfection stability under inflammatory conditions directly determines therapeutic efficacy [63]. We therefore systematically evaluated the cellular uptake of Cy5-mRNA@VE-CB and transfection performance of mLuc@VE-CB under both normal and inflammatory conditions. In terms of cellular uptake, consistent with our previous findings, inflammation impaired the efferocytosis capacity of macrophages (Fig. 4A). Under LPS-induced inflammatory conditions, the uptake rate of VE-CB was slightly reduced. However, inflammation did not affect transfection efficiency of mLuc@VE-CB, suggesting that our vector has the potential to alleviate inflammatory responses. Furthermore, the signal intensity of EGFP was linearly correlated with the dosage of mEGFP administered (Fig. S3D). Under non-inflammatory conditions, VE-CB showed excellent transfection efficiency in both MLE-12 cells and RAW264.7 cells comparable to VE-LNP, highlighting the unique advantage of pH-responsive cubic phase over conventional lamellar vesicle [64]. Critically, in LPS-induced inflammatory cell models, mEGFP@VE-CB retained robust transfection, confirming stable delivery capability in the ARDS-mimetic inflammatory microenvironment (Fig. 4B and 4C). *In vitro* expression kinetics of mIL10@VE-CB were monitored by ELISA (Fig. 4D). It exhibited time-dependent secretion of IL-10 in both MLE-12 cells and RAW264.7 cells, peaking by 24 h, and sustained secretion thereafter (Fig. S3E). This rapid onset aligns with the acute-phase demands of pulmonary inflammation [65]. At 24 h, mIL-10@VE-CB yielded significantly higher IL-10 levels than mIL-10@VE-LNP (Fig. 4E), corroborating the superior mRNA delivery capacity of VE-CB.

Given that VE-L in VE-CB may possess intrinsic antioxidant activity, we assessed the free radical scavenging capacity of VE-L-loaded vectors. In Fig. S4A, our results demonstrated that VE-L effectively eliminated ABTS radicals in a concentration-dependent manner, achieving an inhibition rate of up to 70% at the highest tested concentration [66]. In the context of ARDS, LPS-stimulated macrophages generate excessive reactive oxygen species (ROS), which in turn triggers massive release of inflammatory cytokines. Accordingly, we evaluated whether mIL-10@VE-CB transfection and the vector itself could ameliorate oxidative stress (Fig. 4F). We found that mIL-10@VE-CB significantly reduced intracellular ROS levels as early as 4 h post-transfection and maintained over 60% scavenging

efficiency for up to 24 h in MLE cells and RAW264.7 cells (Fig. 4G, S4B and S4C). In Fig. S4D, mIL-10@VE-CB attenuated the sustained release of pro-inflammatory signals and significantly reduced the secretion levels of key pro-inflammatory cytokines including TNF- $\alpha$ , IL-6, and MCP-1 [67]. Collectively, the synergistic antioxidant action of VE-CB, combined with the anti-inflammatory function of IL-10, holds promise for effectively suppressing oxidative stress in the inflammatory pulmonary microenvironment [68].

Efferocytosis, the process by which phagocytes engulf and clear apoptotic neutrophils during acute inflammatory pathological conditions, plays a pivotal role in promoting inflammation resolution [69, 70]. Our preceding data have confirmed that inflammation exerts a detrimental effect on macrophage efferocytic capacity (Fig. 4A). Therefore, effective restoration of macrophage efferocytosis holds significant implications for inflammatory suppression. We labeled apoptotic neutrophils with fluorescent markers and subsequently evaluated whether mIL-10@VE-CB treatment could ameliorate the efferocytic ability of inflammatory macrophages toward apoptotic neutrophils (Fig. 4H). As shown in Fig. 4I and 4J, LPS-induced inflammatory BMDMs displayed significantly lower uptake of DiD-labeled neutrophils than normal BMDMs, while the mIL-10@VE-CB group restored uptake to near-normal levels [71]. These results further confirm that mIL-10@VE-CB reverses inflammation-induced efferocytosis impairment and promotes apoptotic neutrophil clearance through efficient IL-10 mRNA delivery. Collectively, mIL-10@VE-CB exhibits favorable biocompatibility and synergistic modulation of inflammation, oxidative stress, and macrophage efferocytosis supporting its potential as a promising therapeutic platform for ARDS.



**Fig. 4. *In vitro* transfection efficacy analysis of VE-CB.** (A) Cellular uptake of Cy5-mRNA@VE-CB (left) and transfection performance of mLuc@VE-CB (right) on RAW 264.7 cells under normal and LPS-induced inflammatory conditions ( $n = 3$ ). Transfection efficiency of mEGFP in different formulations under (B) normal or (C) LPS-induced inflammatory condition in MLE-12 cells and RAW264.7 cells, determined by FCM ( $n = 3$ ). (D) Schematic diagram of detecting the *in vitro* expression level of mIL-10 by ELISA (created with BioRender.com). (E) Transfection efficiency of mIL-10@VE-CB in MLE-12 cells and RAW264.7 cells ( $n = 3$ ). (F) Mechanism illustration of reactive oxygen species scavenging by mIL-10@VE-CB: synergistic effect of IL-10 and vitamin E (created with BioRender.com). (G) Time dependence of cellular ROS scavenging rate (4, 12, and 24 h) in MLE-12 cells and RAW264.7 cells ( $n = 5$ ). (H) Schematic diagram showing the restoration of apoptotic neutrophil efferocytosis in inflammatory BMDMs treated with mIL-10@VE-CB (created with BioRender.com). (I) CLSM images of BMDMs treated with mIL-10@VE-LNP or mIL-10@VE-CB.



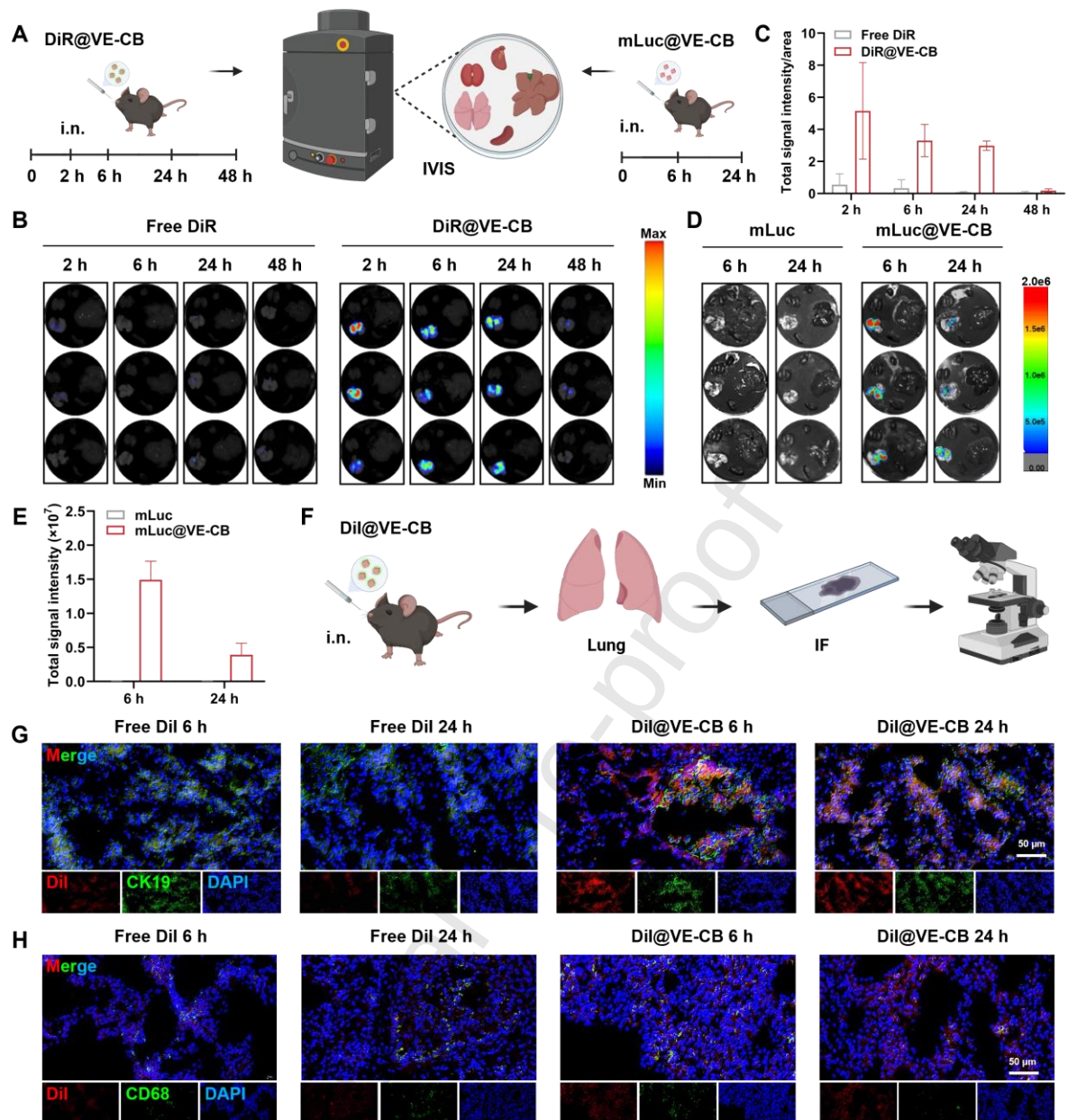
DiD (red, neutrophils); DAPI: (blue, BMDMs). Scale bar: 10  $\mu\text{m}$ . (J) Quantification results of efferocytosis by FCM ( $n = 3$ ). Data were represented as mean  $\pm$  SD. Statistical analysis was conducted using two-tailed unpaired Student's t-test.  $**p < 0.01$ ,  $***p < 0.001$ , *ns* means no significance.

## 2.4 *In vivo* lung distribution and transfection efficiency of VE-CB

An ideal mRNA delivery system for ARDS must combine efficient lung accumulation with sustained local expression [72, 73]. We therefore conducted a comprehensive evaluation of the *in vivo* distribution and pulmonary transfection efficacy of VE-CB (Fig. 5A). Following intranasal administration (*i.n.*) of DiR-labeled VE-CB (DiR@VE-CB), in Fig. 5B, ex vivo imaging revealed pronounced and prolonged lung accumulation [74], with no detectable signal in heart, liver, spleen and kidney at any time point. Quantitative analysis showed that the accumulated fluorescent signal of DiR@VE-CB in the lungs decayed slowly over time, returning to baseline only at 48 h post-administration, significantly longer than that of free DiR (Fig. 5C). This confirmed that VE-CB could effectively prolong pulmonary retention while minimized systemic exposure. To verify the *in vivo* transfection efficiency of VE-CB, we *i.n.* administered mLuc@VE-CB (loaded with Luc mRNA). Luciferase activity peaked at 6 h post-administration and could remained detectable up to 24 h (Fig. 5D and 5E), demonstrating rapid and durable mRNA expression within the lung.

The primary functional targets of inhaled mRNA therapeutics are alveolar type II (AT-II) cells and pulmonary macrophages [75]. Using immunofluorescence on lung sections, we colocalized DiI-labelled VE-CB (DiI@VE-CB) with AT-II cells marker cytokeratin 19 (CK19) or macrophage marker CD68, respectively (Fig. 5F). Quantitative results consistently showed that DiI@VE-CB group exhibited significantly higher overlap of DiI@VE-CB with both CK19 and CD68 (Fig. 5G and 5H) than observed for the control formulations. These findings demonstrated that VE-CB can be efficiently internalized by the two major functional cell types in lung, laying a foundation for its subsequent role in mediating targeted mRNA delivery to pulmonary parenchyma and immune cells alike.





**Fig. 5. *In vivo* distribution and transfection efficiency of VE-CB.** (A) Schematic diagram illustrating the experimental design for evaluating the *in vivo* distribution and transfection efficacy of VE-CB (created with BioRender.com). (B) Fluorescence imaging of major organs at different time points ( $n = 3$ ). (C) Quantitative analysis of fluorescence signal attenuation over time in the lungs of mice intranasally administered DiR@VE-CB or free DiR. (D) Bioluminescence images of major organs and (E) quantitative analysis of luciferase activity at different time points after intranasal administration of free mLuc or mLuc@VE-CB ( $n = 3$ ). (F) Schematic of section analysis of DiI@VE-CB in lung tissues (created with BioRender.com). Immunofluorescence images of DiI@VE-CB (red) colocalized with (G) lung AT-II cells (CK19, green) or (H) macrophages (CD68, green). Scale bar: 50  $\mu\text{m}$ . Data were represented as mean  $\pm$  SD.

## 2.5 *In vivo* therapeutic efficacy of mIL-10@VE-CB in LPS-induced ARDS mice

To evaluate the anti-inflammatory effect of mIL-10@VE-CB, an ARDS mouse model was

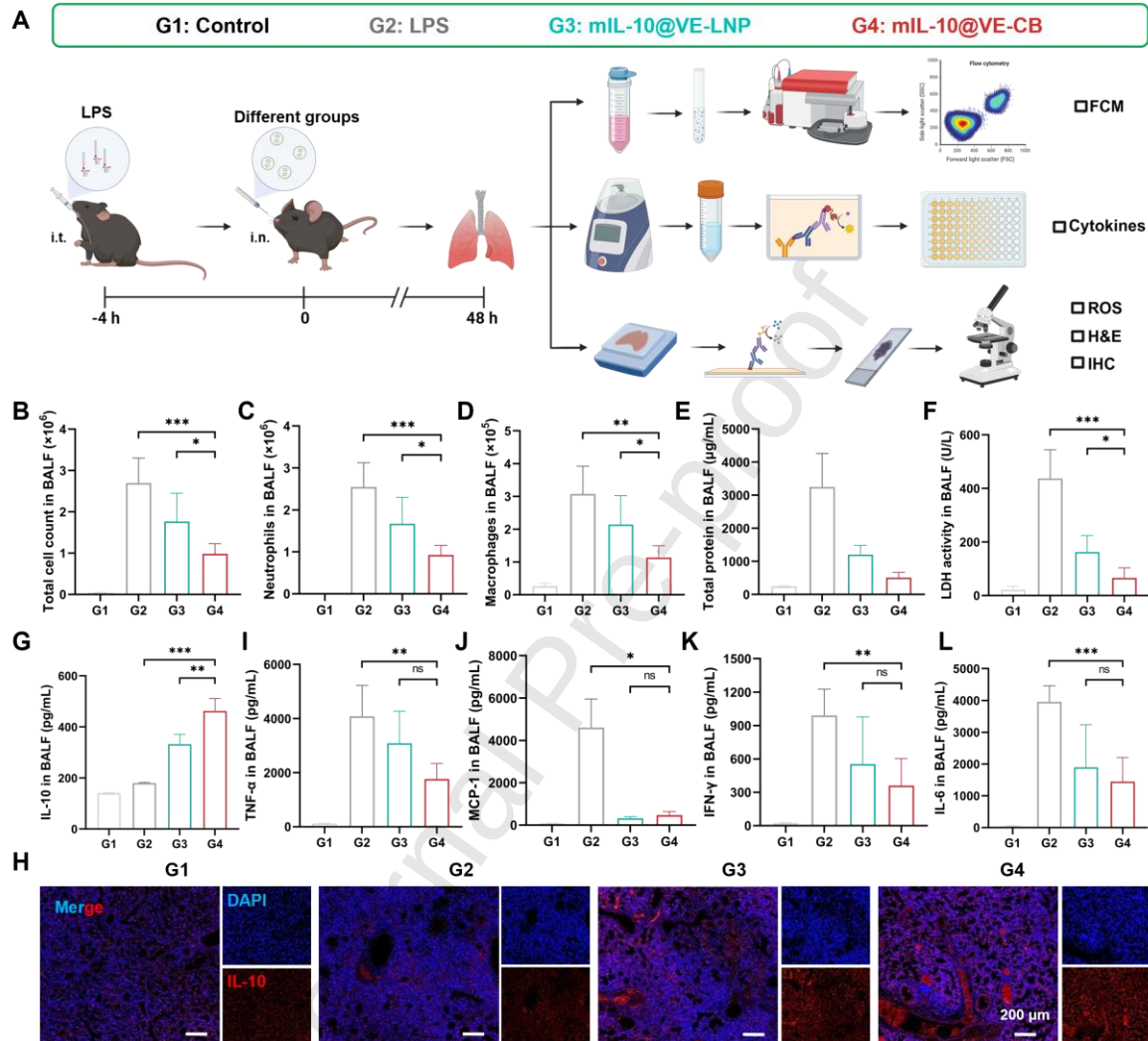
established *via* intratracheal instillation of LPS. Mice were intranasally administered mL-10@VE-LNP or mL-10@VE-CB at a dose of 5 µg mL-10 per mouse. Administration of the formulations was performed 4 h after ARDS model establishment, and 48 h later, bronchoalveolar lavage fluid (BALF) and lung tissues were collected for analytical and pathological examinations (Fig. 6A).

During the progression of ARDS, neutrophils rapidly migrate from peripheral blood to the lungs upon stimulation by inflammatory signals as the first-line immune cells [76]. They exacerbate lung injury by releasing molecules such as myeloperoxidase (MPO). Additionally, macrophages amplify the inflammatory response by phagocytosing pathogens and secreting pro-inflammatory factors [77]. Analysis of inflammatory cell infiltration in BALF is a core indicator for evaluating the severity of ARDS (Fig. S5A). Relative to the LPS group, mL-10@VE-CB group showed a 63% reduction in total BALF cell count, a 64% decrease in neutrophils, and a 63% decline in macrophages (Fig. 6B-D). Moreover, all these metrics in mL-10@VE-CB group were significantly better than those in mL-10@VE-LNP groups. These results indicated that mL-10@VE-CB enabled more efficient inhibition of inflammatory cell recruitment and infiltration by delivering functional IL-10 to the lungs.

Furthermore, sustained inflammatory responses disrupt the integrity of the pulmonary capillary endothelial barrier, leading to increased vascular permeability and subsequent leakage of proteins into the alveolar lumen [78]. As shown in Fig. 6E, the LPS group exhibited a significant increase in protein levels, indicating severe damage to the pulmonary capillary barrier. In contrast, protein levels decreased following administration of mL-10@VE-CB. This change demonstrated that mL-10@VE-CB can alleviate damage to pulmonary capillary endothelial cells and restore vascular barrier integrity by inhibiting inflammatory responses. Lactate dehydrogenase (LDH) activity assay in BALF provided additional insights into alveolar epithelial barrier integrity. The LPS group displayed elevated LDH activity due to widespread alveolar type cell (AT cell) damage, but mL-10@VE-CB group exhibited an 85% reduction in LDH activity relative to the LPS group (Fig. 6F). These confirmed that mL-10@VE-CB better protected AT cells, reduced cell lysis, and mitigated barrier disruption by enhancing IL-10 expression.

Under inflammatory conditions, pulmonary immune cells and parenchymal cells secrete large amounts of various cytokines. To explore this further, we analyzed cytokine expression in BALF. Firstly, to confirm whether VE-CB can efficiently deliver mL-10 and drive its targeted expression in lung, we measured the concentration of IL-10 in BALF. Results showed that IL-10 in BALF of the LPS group was extremely low, indicating that the intrinsic anti-inflammatory mechanism could no longer compensate under ARDS conditions. In contrast, IL-10 levels in mL-10@VE-CB group increased significantly to 462 pg/mL, which was not only much higher than that in the LPS group but also significantly higher than mL-10@VE-LNP group (Fig. 6G). Immunofluorescence section analysis yielded consistent results (Fig. 6H), with distinct IL-10-positive signals observed in lung tissues of mL-10@VE-CB group. These findings clearly confirm that mL-10 is accurately delivered and efficiently translated into functional protein, providing sufficient anti-inflammatory signals to subsequent inhibit the cytokine storm. On this basis, we detected several key pro-inflammatory

cytokines (TNF- $\alpha$ , MCP-1, IFN- $\gamma$ , and IL-6) in BALF using a cytometric bead array (CBA) kit. As shown in Fig. 6I-L, the pro-inflammatory cytokines of the LPS group were significantly increased. In comparison, mL-10@VE-CB group effectively reduced the synthesis and release of these pro-inflammatory cytokines, alleviated the cytokine storm, and regulated the pulmonary microenvironment.



**Fig. 6. ARDS therapy by mL-10@VE-CB treatment.** (A) Experimental schedule of the ARDS model study (created with BioRender.com). (B) Total cell count, (C) neutrophil count and (D) macrophage count in BALF. (E) Protein content in BALF. (F) LDH activity in BALF. (G) IL-10 level in BALF ( $n = 5$ ). (H) CLSM images of IL-10 expression in lung sections after different treatment. IL-10 antibody (red, IL-10); DAPI (blue, nucleus). Scale bar: 200  $\mu\text{m}$ . Levels of pro-inflammatory cytokines including (I) TNF- $\alpha$ , (J) MCP-1, (K) IFN- $\gamma$ , and (L) IL-6 in BALF after treatments ( $n = 5$ ). Data were represented as mean  $\pm$  SD. Statistical analysis was conducted using one-way ANOVA. \* $p < 0.05$ , \*\* $p < 0.01$ , \*\*\* $p < 0.001$ , *ns* means no significance.

## 2.6 Pulmonary injury repair and mechanism of mL-10@VE-CB

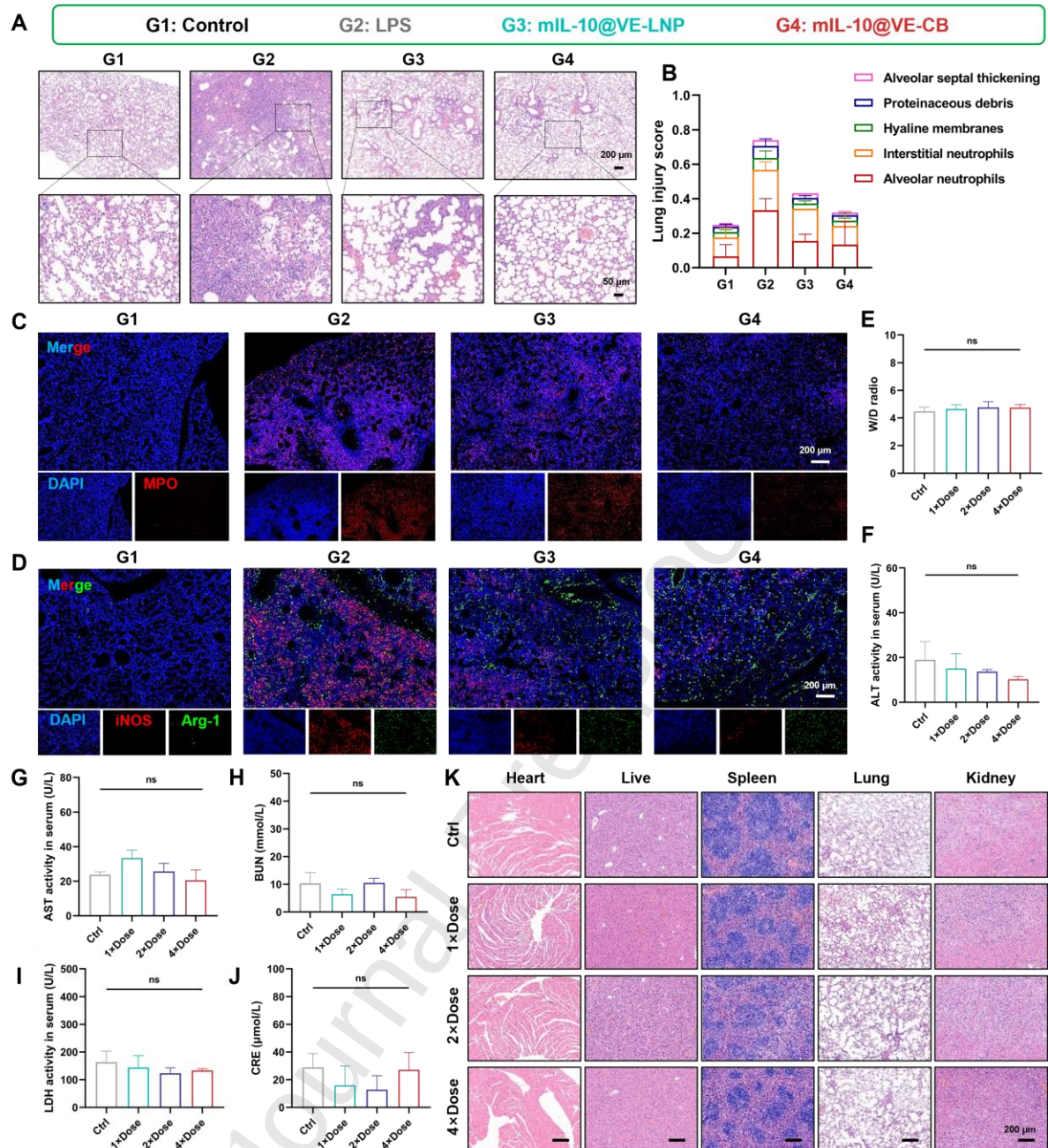
To visually evaluate improvements in lung pathological damage post-treatment, lung tissues from each group were isolated for H&E staining. The LPS group exhibited typical ARDS pathological

features. While lung tissues in mIL-10@VE-CB group showed near-normal pathological morphology, with intact and clear alveolar structures, essentially restored alveolar septal thickness, significantly reduced inflammatory cell infiltration and congestion/edema, and almost complete absence of hyaline membranes (Fig. 7A). Further lung injury scoring demonstrated that mIL-10@VE-CB group had significantly lower scores than other groups (Fig. 7B and S5B), fully confirming that mIL-10@VE-CB delivery effectively alleviates lung pathological damage in ARDS mice. Additionally, MPO immunofluorescence staining further verified neutrophil infiltration. The LPS group displayed elevated LDH activity due to widespread AT cell damage. While mIL-10@VE-CB group displayed significantly attenuated MPO signals with only faint fluorescence in localized areas, confirming its potent inhibitory effect on neutrophil infiltration (Fig. 7C and S5C).

As core regulators of pulmonary inflammation and repair, macrophages exhibit high plasticity differentiating into pro-inflammatory M1 (activated by LPS, marked by iNOS) or anti-inflammatory M2 (induced by IL-10, marked by Arg-1) subtypes based on microenvironmental signals. Immunofluorescence staining of lung sections was used to evaluate the effect of mIL-10@VE-CB on macrophage polarization (Fig. 7D). In the LPS group, lung tissues showed prominent M1 polarization: widespread strong iNOS signals (red) and weak local Arg-1 signals indicating excessive pro-inflammatory macrophage activation. In contrast, the mIL-10@VE-CB group displayed a significant shift toward M2 polarization attenuated iNOS fluorescence and enhanced Arg-1 signals. While mIL-10@VE-LNP also modestly improved polarization balance, their effects were weaker than mIL-10@VE-CB. These results confirm that mIL-10@VE-CB efficiently delivers and expresses mIL-10 in the lung, providing sufficient anti-inflammatory signals to rebalance macrophage polarization, thereby alleviating inflammation and promoting lung repair.

Intranasal administration of mIL-10@VE-CB effectively mitigates LPS-induced ARDS in mice by enhancing pulmonary IL-10 expression, inhibiting inflammation, repairing lung barriers, modulating macrophage polarization, and outperforming mIL-10@VE-LNP.





**Fig. 7. Analysis of H&E-stained and immunofluorescence sections by mIL-10@VE-CB treatment.** (A) H&E staining images of lung sections. Scale bar: 50  $\mu$ m. (B) Lung injury scores based on H&E-stained section observations ( $n = 3$ ). Immunofluorescence analysis of (C) myeloperoxidase (MPO, neutrophil marker) and (D) macrophage phenotypes (iNOS, M1 marker; Arg-1, M2 marker) in lung tissues. Scale bar: 200  $\mu$ m. (E) W/D ratio of lungs treated with different doses ( $n = 3$ ). Blood biochemical parameters including (F) ALT, (G) AST, (H) BUN, (I) LDH, and (J) CRE ( $n = 3$ ). (K) H&E-stained sections of main organs. Scale bar: 200  $\mu$ m. Data were represented as mean  $\pm$  SD. Statistical analysis was conducted using one-way ANOVA. *ns* means no significance.

## 2.6 *In vivo* safety evaluation and *in vitro* long-term storage stability

For pulmonary delivery systems, it is necessary to verify safety at two levels, including local tolerance of lung tissue and systemic off-target effects. To this end, a normal saline control group, a

standard-dose group (1×dose group; 5 µg mRNA per mouse), medium-dose group (2×dose group), and a high-dose group (4×dose group) were established to systematically evaluate the *in vivo* safety of VE-CB. Lung wet to dry (W/D) ratios were indistinguishable across groups, excluding VE-CB-induced alveolar edema or fluid retention, thus avoiding exacerbation of lung damage (Fig. 7E). Further detection of systemic toxicity indicators confirmed that levels of alanine aminotransferase (ALT), aspartate aminotransferase (AST, reflecting liver function), blood urea nitrogen (BUN), creatinine (CRE, reflecting renal function), and lactate dehydrogenase (LDH, reflecting cell damage) in mouse serum of all groups were within the normal physiological range, with no abnormal fluctuations observed as the dose increased (Fig. 7F-J). This suggested that VE-CB had no obvious hepatotoxicity, nephrotoxicity, or risk of systemic cell damage. H&E results in main organs revealed no significant inflammatory infiltration, cell necrosis, or structural abnormalities in tissue sections of the lung, heart, liver, spleen, or kidney in each dose group (Fig. 7K). Specifically, the lung tissue showed intact alveolar structures without alveolar septal thickening or inflammatory cell aggregation. Collectively, these results demonstrated that VE-CB exhibited excellent *in vivo* safety at both effective therapeutic doses and high doses.

To further evaluate the practical application potential of VE-CB for mRNA delivery, we simulated two conditions at 4°C, namely the physiological fluid environment (DMEM medium containing 10% FBS) and the storage buffer environment (PBS, pH 7.4), and determined the encapsulation stability, particle size stability, and transfection activity of VE-CB following storage for different days. Agarose gel electrophoresis and quantitative analyses (Figs. S6A and S6B) showed that encapsulation efficiency only slightly decreased but still maintained above 75% after 30 days. Moreover, the stability of the PBS group was superior to that of the FBS group, confirming that VE-CB exerts a sustained protective effect on mRNA. VE-CB exhibited stable particle size in PBS during 30-day storage, but showed minor aggregation in FBS owing to protein-PEG non-specific interactions, confirming PBS as its optimal storage buffer (Fig. S6C). A further assessment of long-term transfection activity (Fig. S6D) revealed that the MFI value of VE-CB remained at 79.6% of the initial level after 30 days, demonstrating that VE-CB can stably maintain its biological activity under 4°C storage conditions, which provides critical support for subsequent *in vivo* experiments.

### 3. Conclusions

To address the unmet clinical need for targeted ARDS therapy, this study developed vitamin E-based ionizable lipid-derived virus-mimetic cubosomes for precise pulmonary delivery of mL-10. Its core advantage lied in structure-function synergy. GMO enabled stable cubic phase formation, while ionizable lipid VE-L conferred pH responsiveness. In acidic endosomes (pH 5.0-5.5), this synergy drove a lamellar vesicle-to-cubic phase transition boosting membrane fluidity, enhancing endosomal membrane fusion, and overcoming the key bottleneck of mRNA delivery. VE-CB achieved mRNA transfection efficiency comparable to commercial S-LNP and far superior to traditional VE-LNP, with excellent physicochemical properties that could solve traditional mRNA delivery issues. *In vivo*,



intranasal VE-CB enabled highly efficient lung targeting and efficiently delivery of mL-10 to AT cells and macrophages for sustained expression. In ARDS mice, mL-10@VE-CB significantly reversed pulmonary pathological damage, with excellent biosafety. VE-CB also efficiently delivered diverse nucleic acids and can be extended to other pulmonary diseases or targeted systemic nucleic acid therapies. In conclusion, VE-CB was a safe, efficient, and versatile lung-targeted nucleic acid platform, offering a novel strategy for precise ARDS mRNA therapy, substantial clinical translation potential, and a foundation for advancing mRNA-based precision medicine.

#### 4. Materials and methods

##### Materials

The vitamin E-derived ionizable lipid (VE-L) was developed through independent collaboration within the laboratory. Organic reagents used in this study were purchased from Sinopharm Chemical Reagent Co., Ltd. (Shanghai, China). DL- $\alpha$ -tocopherol was obtained from Aladdin Reagent (Shanghai) Co., Ltd. (Shanghai, China). Glyceryl monooleate (GMO) was purchased from Maya Reagent Co., Ltd. (Shanghai, China). 1,2-Distearoyl-sn-glycero-3-phosphoethanolamine-N-[methoxy(polyethylene glycol)-2000] (DSPE-PEG<sub>2000</sub>) was acquired from AVT Pharmaceutical Technology Co., Ltd. (Shanghai, China). Cholesterol (Chol), 1,2-Dioleoyl-sn-glycero-3-phosphoethanolamine (DOPE), 1,2-Dioleoyl-sn-glycero-3-phosphocholine (DOPC), and 1,2-Dipalmitoyl-sn-glycero-3-phosphocholine (DPPC) were purchased from Xi'an Ruixi Biotechnology Co., Ltd. (Xi'an, China). IL-10 mRNA (mIL-10) was obtained from Shanghai Hexincheng Biotechnology Co., Ltd. (Shanghai, China). Cy5-labeled mRNA (Cy5-mRNA) was donated by Shenzhen Ruiji Biotechnology Co., Ltd. (Shenzhen, China). Luciferase mRNA (Luc mRNA, mLuc) was purchased from Nanjing Vazyme Biotech Co., Ltd. (Nanjing, China). Enhanced Green Fluorescent Protein mRNA (EGFP mRNA, mEGFP) and EGFP plasmid DNA (EGFP pDNA, pEGFP) were acquired from GenScript (Nanjing, China). Luciferase siRNA (Luc siRNA, siLuc) was purchased from Sangon Bioengineering (Shanghai) Co., Ltd. (Shanghai, China). RiboGreen RNA Detection Reagent was purchased from Shanghai Maokang Biotech Co., Ltd. (Shanghai, China). The Luciferase Reporter Gene Assay Kit was purchased from Yeasen Biotechnology (Shanghai) Co., Ltd. (Shanghai, China). The Cell Counting Kit-8 (CCK-8) assay kit was purchased from Wuhan Robi Biotechnology Co., Ltd. (Wuhan, China). Lipopolysaccharide (LPS) was obtained from Sigma-Aldrich (St. Louis, MO, USA). The Cytometric Bead Array (CBA) kit for cytokines was purchased from BD Biosciences (New Jersey, USA). The Lactate Dehydrogenase (LDH) kit was obtained from Nanjing Jiancheng Bioengineering Institute (Nanjing, China). The IL-10 ELISA detection kit was purchased from Wuhan Solarbio Science & Technology Co., Ltd. (Wuhan, China). All reagents were used without further purification unless otherwise specified.

##### Cell lines and animals

CT26 cells, MC 38 cells, MLE-12 cells, and RAW 264.7 cells were purchased from the Cell Bank of the Chinese Academy of Sciences. Luciferase-expressing MC38 (MC38-Luc) cells were constructed using the lentiviral method. CT26 cells were cultured in 1640 medium supplemented with 10% fetal

bovine serum (FBS), 1% penicillin/streptomycin(v/v), and glutamine. MLE-12 cells and MC38-Luc cells were cultured in DMEM medium supplemented with 10% FBS and 1% penicillin-streptomycin (v/v). RAW 264.7 cells were cultured in DMEM medium supplemented with 10% FBS.

C57BL/6 mice (male, 6-8 weeks old) were purchased from the Experimental Animal Resource Center of Huazhong University of Science and Technology (HUST). All animals were housed in a specific pathogen-free (SPF) environment, following the guidelines for experimental animal ethics of HUST. All animal experiments were approved by the Institutional Animal Care and Use Committee (IACUC) of HUST (Number: HUST-IACUC-2025-0042).

### **Preparation of VE-CB, VE-LNP, and S-LNP**

VE-CB was prepared via the ethanol injection method. Nucleic acids (EGFP mRNA, IL-10 mRNA, Luc mRNA, Luc siRNA, and EGFP pDNA) were completely dissolved in a 0.2 M sodium acetate buffer at pH 5.5. Lipids including VE-L, GMO, Chol, and DSPE-PEG<sub>2000</sub> were weighed according to the required mass ratio and dissolved in ethanol, followed by mixing at different molar ratios as per experimental formulations. The mass ratio of ionizable lipid to mRNA was maintained at 20: 1. Subsequently, the ethanol solution containing lipids was rapidly injected into the nucleic acid-containing buffer solution for quick mixing, with an ethanol-to-water volume ratio of 1: 5. Gentle pipetting was performed to ensure thorough mixing. Ethanol was removed by dialysis against PBS (pH 7.4) at 4°C overnight using a 10 kDa MWCO dialysis membrane.

For VE-LNP, its formulation does not contain GMO, and the other lipid components are consistent with those of VE-CB. The lipid concentration and mRNA concentration of SM102-LNP (S-LNP) are consistent with those in commercial methods. The resulting mixture was diluted with cell culture medium for *in vitro* cell experiments or with PBS for *in vivo* animal experiments, depending on specific experimental requirements.

### **Physicochemical characterization of VE-CB**

Fresh cubosomes was diluted with enzyme-free water. The particle size, polydispersity index (PDI), and zeta potential of VE-CB were measured using a dynamic light scattering (DLS) instrument (ZetaPALS, Brookhaven Instruments). For morphological observation, 10 µL of diluted VE-CB was dropped onto a copper grid. 1% phosphotungstic acid solution was added for negative staining. The grid was air-dried at room temperature to remove excess moisture, and then VE-CB were observed using a transmission electron microscope (TEM).

### **mRNA encapsulation efficiency**

The encapsulation efficiency was determined by RiboGreen RNA detection reagent. mRNA@VE-CB were treated with an appropriate amount of 2% Triton X-100 to break the emulsion and release the encapsulated mRNA. RiboGreen was added according to the kit instructions, and the mixture was incubated in the dark for 5 min. The fluorescence intensity of each group was measured using a multimode microplate reader ( $\lambda_{ex}/\lambda_{em}$  = 480/520 nm). The encapsulation efficiency was calculated according to the following formula:

Encapsulation Efficiency (EE, %) = (Total amount of RNA – Free RNA amount) / Total amount of

RNA×100%

### **Agarose gel electrophoresis**

1% agarose gel containing GelBlue stain was prepared. Five microliters of VE-CB samples with different lipid/nucleic acid (L/N) mass ratios were mixed with loading buffer and loaded onto the gel. Electrophoresis was performed at 115 V for 40 min, and the bands were visualized using a gel imaging system.

### **Cell viability of VE-CB**

MLE-12 cells and RAW264.7 cells were separately seeded into 96-well plates. After incubation for 24 h (37°C, 5% CO<sub>2</sub>). Different VE-CB concentrations or mRNA concentrations with blank medium were added to the wells via medium replacement. The cells were further incubated for 24 h. Subsequently, 10 µL of CCK-8 reagent was added to each well, and the plates were incubated in an incubator for another 4 h. The absorbance at 450 nm was measured using a microplate reader (Bio-Tek).

### **Hemolysis assay**

Mouse blood was collected, centrifuged, and washed to obtain RBC. The washed RBC were mixed with VE-CB at different concentrations, then incubated in a thermostatic shaker (37°C, 100rpm) for 4 h. After incubation, the mixture was centrifuged. The supernatant was transferred to a 96-well plate, and the absorbance at 542 nm was measured using a microplate reader. The hemolysis Rate was calculated according to the following formula:

$$\text{Hemolysis Rate (\%)} = (\text{Sample absorbance} - \text{Negative control absorbance}) / (\text{Positive control absorbance} - \text{Negative control absorbance}) \times 100\%$$

### ***In vitro* Luc siRNA delivery**

Luciferase-expressing MC38 cells were seeded into 96-well plates and cultured in complete medium containing 10% FBS for 24 h. The original medium was aspirated, and the cells were washed 3 times with serum-free DMEM. Subsequently, VE-CB, VE-LNP, and S-LNP loaded with siLuc were added (siRNA: 1 µg/mL). A negative siRNA control group (unrelated sequence delivered by the same carrier) and a blank control group (serum-free medium only) were also set up. After further incubation for 24 h, the medium was aspirated. 70 µL pre-cooled cell lysis buffer was added to each well, and lysed on ice for 10 min. 30 µL luciferase substrate was then added, and the relative luminescence units (RLU) were measured by microplate reader.

### ***In vitro* EGFP mRNA or EGFP pDNA delivery**

CT26, MLE-12, or RAW264.7 cells were seeded into 24-well plates and cultured for 24 h. The original medium was aspirated, and the cells were washed 3 times with serum-free DMEM. VE-CB, VE-LNP, S-LNP loaded with mEGFP or pEGFP were added, with the final concentration of both mEGFP and pEGFP set to 500 ng/mL. The cells were cultured for 24 h (mEGFP) or 48 h (pEGFP). After incubation, the medium was aspirated, and the cells were washed 3 times with PBS. The cells were then collected and resuspended in PBS. The mean fluorescence intensity (MFI) was detected by flow cytometer (FCM) to evaluate transfection efficiency.

***In vitro* IL-10 mRNA expression**

MLE-12 or RAW264.7 cells were seeded into 24-well plates. mIL-10@VE-LNP and mIL-10@VE-CB were added with serum-free medium. IL-10 mRNA was at a final concentration of 500 ng/mL. The cells were further cultured for 24 h, and then the cell supernatant was collected and resuspended in PBS. The concentration of IL-10 in the supernatant was determined using an enzyme-linked immunosorbent assay (ELISA) according to the kit instructions and were measured by microplate reader.

**Induction of inflammatory cells**

MLE-12 or RAW264.7 cells were seeded into 24-well plates. Lipopolysaccharide (LPS) was added to the wells at a final concentration of 200 ng/mL, and the cells were cultured overnight for 12 h to induce cell activation and inflammatory factor secretion, thereby establishing an inflammatory cell model.

**Analysis of cellular uptake mechanism**

Cells were then treated with different cellular uptake inhibitors: amiloride (macropinocytosis inhibitor), sucrose (clathrin-mediated endocytosis inhibitor), chloroquine (endosome acidification inhibitor), cytochalasin D (actin polymerization inhibitor), methyl- $\beta$ -cyclodextrin (M- $\beta$ -CD, caveolin-mediated endocytosis inhibitor), and 4°C incubation (energy-dependent process inhibition). VE-CB was labeled with DiD, and after 4 h of administration for uptake, the MFI was detected by FCM. The cellular uptake inhibition rate (CUIR) was calculated.

**CLSM analysis of cellular uptake and endosomal escape**

MLE-12 cells were seeded into confocal dishes and cultured for 24 h. Subsequently, Cy5-mRNA@VE-CB (Cy5-mRNA: 1  $\mu$ g/mL) was added for 4 h of uptake. Lysosomes were labeled with LysoTracker Green, and cell nucleus were labeled with Hoechst 33342 or DAPI. After each staining step, cells were gently washed 3 times with cold PBS, and imaging was performed by Nikon confocal laser scanning microscope (CLSM). The Pearson colocalization coefficient between Cy5-mRNA and LysoTracker Green was calculated using the Coloc 2 plugin of ImageJ software. Meanwhile, the distribution of Cy5-mRNA was observed to comprehensively evaluate cellular uptake and endosomal escape efficiency.

**Small-angle X-ray scattering (SAXS) measurement**

VE-CB (lipid concentration: 10 mg/mL) was prepared and dispersed in buffers at pH 7.4 and 5.5. The samples were sealed in quartz capillaries (1.5 mm diameter) and equilibrated at 25°C for 30 min before measurement. SAXS measurements were performed using a SAXSpoint 5.0 small-angle X-ray scattering system (Anton Paar, Austria) equipped with a Cu K $\alpha$  X-ray source ( $\lambda = 1.54 \text{ \AA}$ ). The sample-to-detector distance was 500 mm, covering a scattering vector ( $q$ ) range of 0-0.40  $\text{\AA}^{-1}$ , where  $q = 4\pi\sin\theta / \lambda$ . Data acquisition time was 30 min per sample. Background scattering from the buffer was subtracted from all sample patterns.

**Cryogenic TEM (Cryo-TEM) observation**

VE-CB (lipid concentration: 5 mg/mL) were dispersed in buffer adjusted to pH 7.4 or 5.5. Then 5  $\mu$ L aliquot of the sample was applied onto a glow-discharged holey carbon grid. The excess liquid was

gently removed by blotting with filter paper. The grid was immediately transferred to a vitrification device and rapidly frozen in liquid ethane cooled by liquid nitrogen and then transferred to a cryo-holder under liquid nitrogen cooling. Imaging was performed using an FEI Tecnai Spirit cryogenic TEM (Thermo Fisher Scientific, USA) operated at an accelerating voltage of 120 kV.

#### **Verification of VE-CB membrane fluidity**

VE-CB (lipid concentration: 10 mg/mL) was adjusted to pH 7.4 and 5.5 respectively. 1,6-diphenyl-1,3,5-hexatriene (DPH) fluorescent probe was added at a final concentration of 5  $\mu$ M, and the mixture was incubated at 37°C in the dark for 30 min. Each sample to be tested was then added to a 96-well plate, and the fluorescence intensity at 430 nm was detected by multi-functional microplate reader. Simultaneously, the absorbance at 400 nm was measured to assist in verifying changes in membrane fluidity.

#### **Membrane fusion ability of VE-CB**

The mice RBC suspension was prepared according to the aforementioned hemolysis assay method. The suspension was mixed with VE-CB (lipid concentration: 10 mg/mL) at pH 7.4 and 5.5. After centrifugation, the absorbance at 450 nm was detected to calculate the hemolysis rate. The RBC suspension was washed with PBS, centrifuged at 40,000 rpm for 10 min to discard the supernatant, dissolved in 40 $\times$  volume of 0.25 $\times$  PBS, and placed at 4°C for 2 h for sufficient hemolysis. After centrifugation and further washing with PBS, the pink pellet obtained was RBC-derived nanovesicles (NV, simulating endosomal membrane). VE-CB was labeled with DiI ( $\lambda_{ex}/\lambda_{em}$  = 549/565 nm), and NV was labeled with DiD ( $\lambda_{ex}/\lambda_{em}$  = 644/663 nm). DiI@VE-CB and DiD@NV were mixed under pH 7.4 and pH 5.5 conditions, respectively, and incubated at 37°C for 40 min. The fluorescence intensity ranging from 550 nm to 700 nm at excitation wavelength of 525 nm. The FRET efficiency was calculated based on the ratio of DiD fluorescence intensity to DiI fluorescence intensity. For the FRET pair DiI and DiO ( $\lambda_{ex}/\lambda_{em}$  = 484/501 nm), DiI was used to label VE-CB, while DiO was used to label NV.

#### ***In vivo* transfection efficiency**

The mLuc@VE-CB, mLuc@VE-LNP, and mLuc@S-LNP were prepared (5  $\mu$ g mLuc per mouse) with an administration volume of 40  $\mu$ L per mouse. The Ctrl group (normal saline) was administered an equal volume of normal saline. All formulations were delivered intranasally. At 6 h and 24 h post-administration, mice were intraperitoneally (i.p.) injected with D-Glutamine in potassium salt at a dose of 150 mg/kg. Subsequently, mice were sacrificed, and major organs (heart, liver, spleen, lung, and kidney) were rapidly isolated. Each organ was sequentially placed in an *in vivo* imaging system (IVIS, Pearl Trilogy, LI-COR) for bioluminescent imaging, and the fluorescence intensity of each organ was quantified to evaluate transfection efficiency.

#### ***In vivo* distribution of VE-CB**

The DiR@VE-CB and Free DiR were prepared (5  $\mu$ g DiR per mouse) with an administration volume of 40  $\mu$ L per mouse and the administration route is intranasal instillation (*i.n.*). At 2, 6, 24, and 48 h post-administration, mice were sacrificed, and major organs (heart, liver, spleen, lung, kidney)



were quickly isolated. The organs were rinsed with pre-cooled PBS to remove residual blood, and surface moisture was blotted dry with filter paper. Each organ was then imaged using an *in vivo* imaging system (IVIS, Pearl Trilogy, LI-COR) for fluorescence imaging. The fluorescence intensity of each organ was quantified to calculate two parameters: the fluorescence proportion of each organ and the temporal change in lung fluorescence intensity. Lung tissues were fixed with 4% paraformaldehyde, stored at 4°C. We performed immunofluorescence staining using AT-II cells marker CK19 or macrophage marker CD68, and observed the results via CLSM.

#### **Intracellular ROS detection**

RAW264.7 cells were seeded into 96-well plates and cultured for 24 h. Except for the control (Ctrl) group, all other groups were first treated with LPS (200 ng/mL) and incubated overnight for 12 h to induce ROS production. The cells were treated with blank VE-CB or mIL-10@VE-CB (500 ng/mL IL-10 mRNA), followed by a 24h incubation. After incubation, the medium was aspirated, and cells were gently washed 3 times with serum-free DMEM. Then, 20  $\mu$ M DCFH-DA was added to each well, and the cells were incubated at 37°C in the dark for 30 min. Residual extracellular probe was removed by washing 3 times with serum-free DMEM. Finally, the intracellular fluorescence intensity ( $\lambda_{ex}/\lambda_{em} = 488/525$  nm) was detected using a microplate reader to quantify ROS levels.

#### ***In vitro* efferocytosis assay for macrophage phagocytosis of apoptotic neutrophils**

Bone marrow-derived macrophages (BMDMs) were isolated from C57BL/6 mice and cultured in DMEM medium containing 10% FBS, 1% penicillin-streptomycin, and 20 ng/mL M-CSF for 7 days to induce differentiation (37°C, 5% CO<sub>2</sub>). Neutrophils were purified from mouse bone marrow, washed twice with PBS, labeled with 5  $\mu$ M DiD dye at 37°C. in the dark for 30 min, and washed 3 times to remove free dye. Apoptosis of DiD-labeled neutrophils was induced by UV irradiation (254 nm) for 30 min. BMDMs were seeded into 24-well plates and stimulated with LPS to establish an inflammatory model, followed by addition of mIL-10@VE-CB, mIL-10@VE-LNP, or blank medium and cultured for 24 h. DiD-labeled apoptotic neutrophils were added to BMDMs cultures at a macrophage-to-neutrophil ratio of 1:4 and incubated for 2 h at 37°C. Non-phagocytosed neutrophils were removed by gentle PBS washing 3 times, and the phagocytic efficiency of BMDMs was quantified by FCM based on DiD MFI or observed and analyzed by CLSM.

#### **Establishment of the ARDS model**

Male C57BL/6 mice (6-8 weeks, weighing 20-22 g) were used to establish the ARDS model via intratracheal instillation of LPS. The mice were anesthetized with an intraperitoneal injection of tribromoethanol (250 mg/kg). A tracheal atomization needle (24G) was slowly inserted into the trachea, and LPS (dissolved in PBS) was rapidly instilled at a dose of 5 mg/kg. After instillation, the mice were suspended vertically for 30 s to ensure uniform distribution of LPS in the lungs. Mice in the Ctrl group (healthy mice) received no LPS instillation. The mice were then placed on a heating pad until they regained consciousness.

#### **Evaluation of therapeutic efficacy**

The method for establishing the ARDS model was the same as described previously. Mice were

randomly divided into 4 groups: the Ctrl group (healthy mice), LPS group, mL-10@VE-LNP group and mL-10@VE-CB group. All treatment groups received intranasal instillation (*i.n.*), with a uniform dose of 5 µg mL-10 per mouse. 48h after LPS intervention, bronchoalveolar lavage fluid (BALF) and lung tissues were collected from each group for subsequent analysis.

For cytokines in BALF, detection was performed using a mouse inflammatory CBA kit (BD Biosciences, USA), and data analysis was conducted via FCM. IL-10 was quantified separately using a specific ELISA kit. Lung tissues were fixed with 4% paraformaldehyde, stored at 4°C, embedded in paraffin to prepare sections, and hematoxylin-eosin (H&E) staining was used to observe pulmonary pathological damage. The degree of lung tissue damage in each group was scored blindly according to the Matute-Bello lung injury scoring standard, and the average score of each group was calculated.

Immunofluorescence staining was used to analyze the expression of different indicators in lung tissues. IL-10 expression and distribution in lung tissues were observed using CLSM after labeling with an IL-10 antibody. Myeloperoxidase (MPO) antibody was used for staining. The expression in lung tissues was observed via CLSM and fluorescence intensity was quantified using ImageJ. M1 macrophages were labeled with an inducible nitric oxide synthase (iNOS) antibody, and M2 macrophages were labeled with an arginase-1 (Arg-1) antibody. The expression of these two markers was observed by CLSM, and quantification was also performed using Image J.

### ***In vivo* safety evaluation**

Mice were randomly divided into 4 groups: Ctrl group (normal saline), 1× dose group (normal dose: 5 µg IL-10 mRNA per mouse), 2× group (medium-dose: 10 µg IL-10 mRNA per mouse), and 4× group (high-dose: 20 µg IL-10 mRNA per mouse). All groups were administered via intranasal instillation (*i.n.*). At 24 h post-administration, whole lungs and serum were collected. The wet-to-dry (W/D) lung weight ratio was calculated by dividing the wet lung weight by the dry lung weight (after drying to constant weight). A series of serum biochemical indicators, including ALT, AST, BUN, LDH, and CRE, were detected to evaluate liver and kidney function. Additionally, major organs (heart, liver, spleen, lung, kidney) were excised, fixed, embedded in paraffin, and sectioned for H&E staining. Histopathological changes in the tissues were observed under a light microscope.

### **Supporting information**

Supporting information of this article can be found online.

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## Competing interests

All authors declare no conflict of interest.

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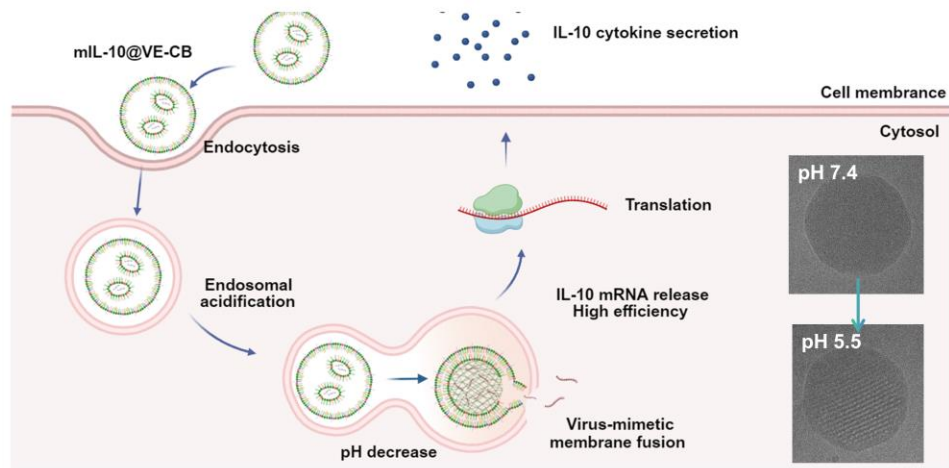
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## Highlights

1. Virus-mimetic cubosomes (VE-CB) enable cytosolic delivery of IL-10 mRNA.
2. VE-CB undergoes pH-triggered lamellar-to-cubic phase transition for endosomal escape.
3. Intranasal VE-CB achieves sustained pulmonary IL-10 expression in ARDS mice.
4. A versatile platform for precision pulmonary mRNA therapy with translational potential.